

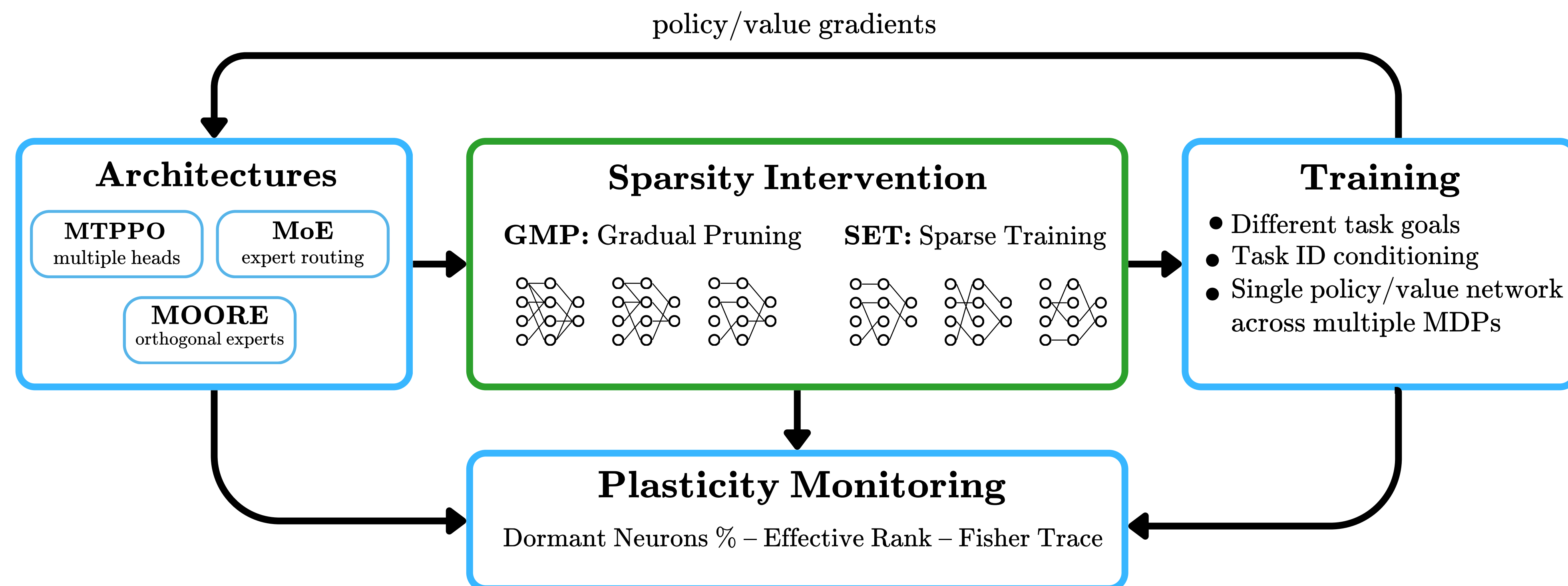
Sparsity-Driven Plasticity in Multi-Task Reinforcement Learning



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Motivation

- Deep RL agents often suffer **plasticity loss**: they stop adapting as training progresses.
- In **multi-task RL**, plasticity is even more critical – one agent must flexibly solve diverse, sometimes conflicting tasks.
- **Dynamic sparsity** (via pruning or sparse training) is a simple, general mechanism to preserve plasticity and improve both adaptability and efficiency.

Key Contributions

- **Dynamic sparsity preserves plasticity**: Gradual Magnitude Pruning (GMP) and Sparse Evolutionary Training (SET) reduce neuron dormancy, maintain higher representation rank, and stabilize Fisher trace.
- **Plasticity gains align with performance**: sparse agents often outperform dense baselines and rival explicit plasticity methods.
- **Architecture matters**: gains are strongest in MTPPO and MoE, with more nuanced effects for MOORE.

Benchmarks

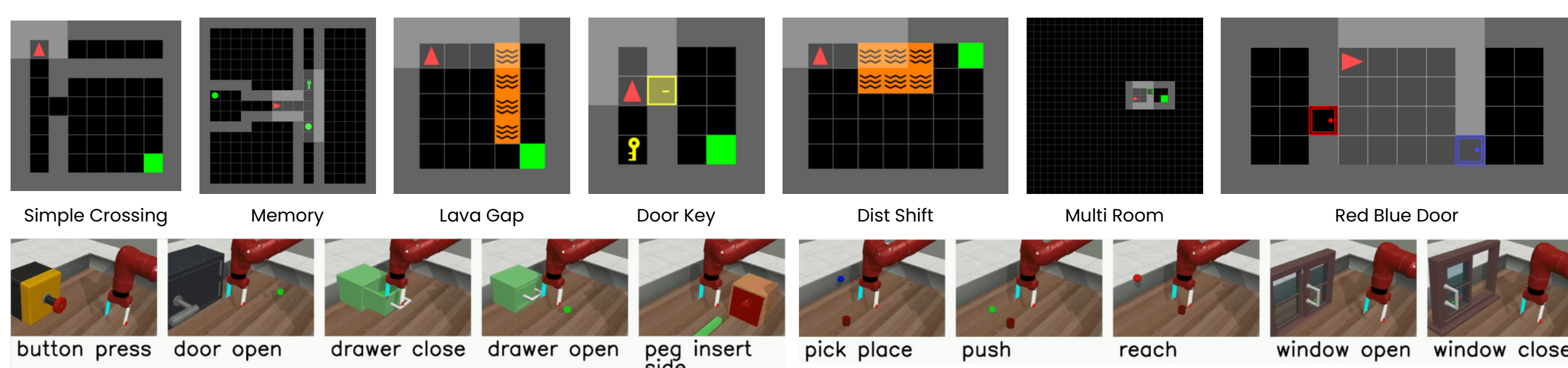


Figure 1: Multi-Task Benchmarks. MiniGrid (top) for discrete navigation, and Meta-World (bottom) for continuous robotic manipulation.

Plasticity metrics

- **Dormant neuron %**: $\beta_\tau = \frac{\sum_{l \in L_{\text{all}} \setminus \{L_{\text{out}}\}} H_\tau^l}{\sum_{l \in L_{\text{all}} \setminus \{L_{\text{out}}\}} H^l}$
- **Effective Rank**: $\text{srnk}_\delta(\Phi) = \min_k \left\{ \frac{\sum_{i=1}^k \sigma_i}{\sum_{i=1}^d \sigma_i} \geq 1 - \delta \right\}$.
- **Fisher Information Trace**: $\text{Tr}(F) = \mathbb{E}_{s, a \sim \pi} [\|\nabla_\theta \log \pi(a|s)\|_2^2]$.

Limitations & Future Work

- **Efficiency**: GMP trains dense networks with masks (no training-time speedup); SET requires sparse kernels for real gains.
- **Sensitivity**: performance depends on sparsity level and hyperparameters – no one-size-fits-all setting.

Publication Details

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Results: Multi-Task Performance

Table 1: Mixture of Experts (MoE) performance across MiniGrid benchmarks. Sparsity interventions (GMP and SET) demonstrate performance better or comparable to strong, explicit plasticity interventions like ReDo. Both approaches substantially outperform the dense baseline, indicating that maintaining plasticity is key for multi-task success.

Agent Treatment	MT3		MT5		MT7	
	IQM (↑)	95% CI	IQM (↑)	95% CI	IQM (↑)	95% CI
<i>Mixture of Experts (MoE)</i>						
Dense	0.74	(0.71, 0.76)	0.77	(0.70, 0.82)	0.80	(0.75, 0.84)
Gradual Pruning	0.77	(0.74, 0.79)	0.84	(0.78, 0.86)	0.87	(0.83, 0.88)
SET	0.76	(0.74, 0.78)	0.79	(0.72, 0.85)	0.82	(0.78, 0.85)
ReDo	0.77	(0.76, 0.80)	0.82	(0.81, 0.85)	0.85	(0.82, 0.88)
Reset	0.64	(0.54, 0.73)	0.78	(0.73, 0.83)	0.84	(0.80, 0.87)
Weight Decay	0.75	(0.71, 0.76)	0.78	(0.70, 0.85)	0.77	(0.71, 0.82)
LayerNorm	0.44	(0.38, 0.46)	0.34	(0.29, 0.40)	0.39	(0.32, 0.43)

Results: Plasticity Profiles

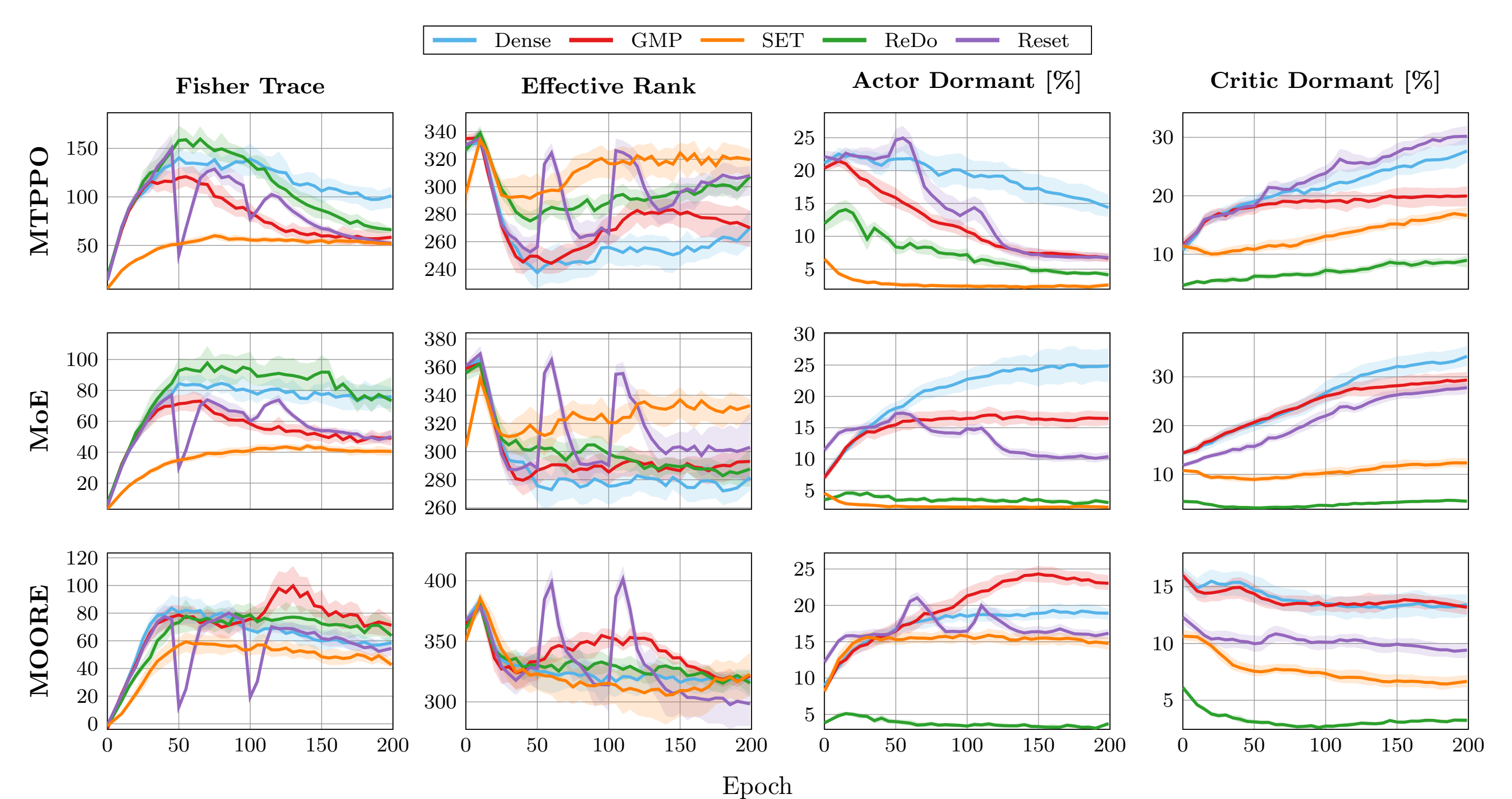


Figure 2: Sparsity vs. Plasticity Interventions. Plasticity dynamics of sparse methods (GMP, SET) versus ReDo, Reset and a dense baseline, across MTPPO, MoE, and MOORE architectures on the MT5 benchmark. Sparse agents often achieve lower dormancy and higher effective rank, often surpassing explicit plasticity-preserving techniques.

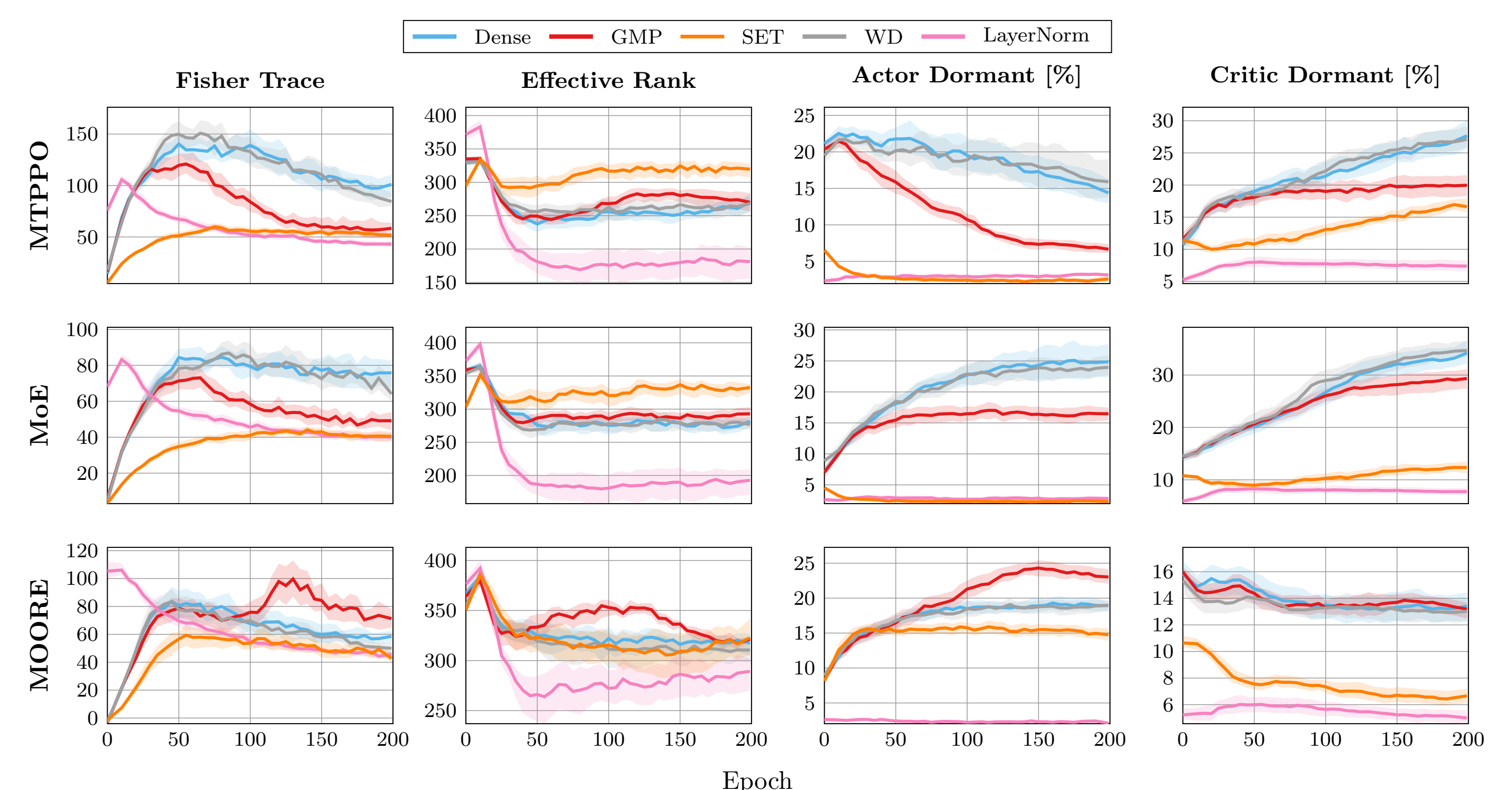


Figure 3: Sparsity vs. Common Regularizers. Compared to dense agents with LayerNorm or Weight Decay, sparse agents (GMP, SET) sustain plasticity more effectively. Sparsity avoids rank collapse from LayerNorm and reduces dormancy more consistently than Weight Decay, leading to stronger adaptability.